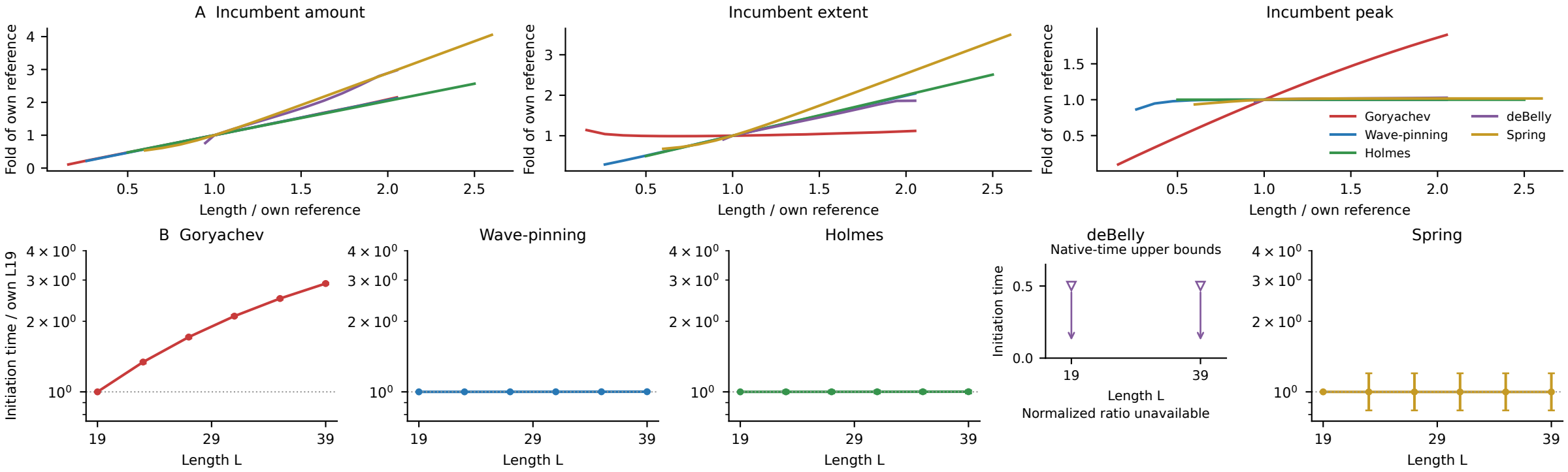
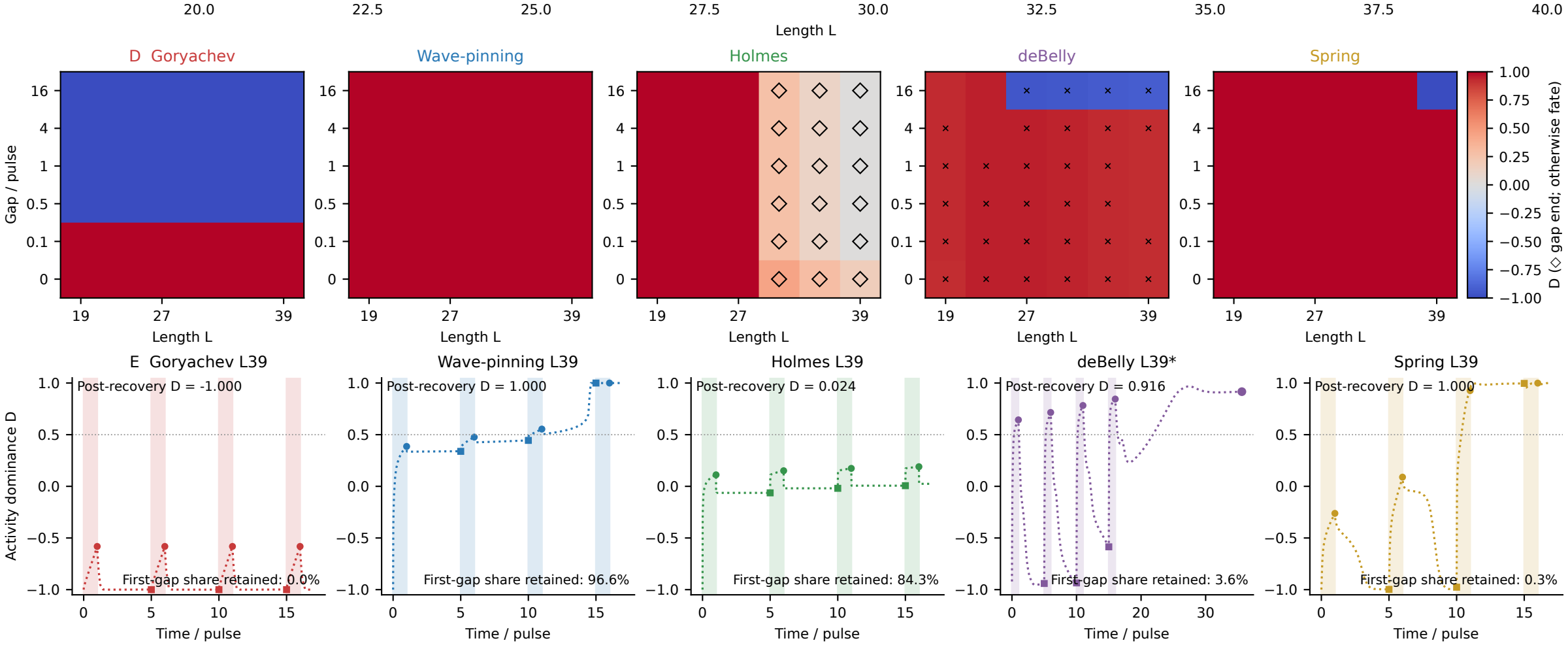
Replacement resistance in enlarged cells arises through three mechanism classes; cue shape and timing distinguish their signatures





F Shape and timing signatures distinguish mechanism classes

Class / models	Initiation slows	Equal-area penalty	Share through gaps	Commit after collapse	Experimental measurement
Initiation-limited Goryachev	Yes (2.90×)	Modest (1.12×)	Resets	No	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Mass-conserved, saturating Wave-pinning, Holmes	No resolved slowing	Large; grows over resolved L range	Share retained (ratchet)	—	Beam width at fixed photon dose; pulse trains; gap-end biosensor share
Slow non-activity state deBelly*, Spring	Unresolved* / no	Modest (~1×)	Collapses early	Yes	Beam width at fixed photon dose; pulse trains; gap-end biosensor share

◇ Holmes L31–L39: last inter-pulse gap-end D, with no target-length calibration. × failed field tail; ? uninformative ratio.

* deBelly initiation is interval-censored; later-gap timing and endpoint fields carry numerical qualifications (complete supplement).